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Deep Learning in Asset Pricing

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Background

Traditional asset pricing models, such as the Capital Asset Pricing Model (CAPM) and Fama–French factor models, seek to explain differences in expected stock returns through exposure to systematic risk. However, these models typically impose restrictive functional forms and rely on a predetermined set of risk factors.

Research Questions

This paper attempts to answer the following: Can deep learning improve the estimation of the stochastic discount factor and explain differences in expected stock returns?

Model Innovations

The model formalizes a deep neural network that estimates an asset pricing model for individual stock returns. This estimation rests on three innovations:

- The fundamental no-arbitrage condition
- An adversial approach for testing assets
- Economics states extracted from macroeconomic time series.

No-Arbitrage Asset Pricing

No-Arbitrage Asset Pricing

Let R_{t+1} denote return of asset i at time $t+1$. The no-arbitrage assumption is the stochastic discount factor (SFD) $M_{t+1} > 0$, for any excess returns $R^e_{t+1,i} = R_{t+1,i} - R^f_{t+1}$, we have

$$\mathbb{E}_t\left[M_{t+1}R^e_{t+1,i}\right] = 0 \iff \mathbb{E}\left[R^e_{t+1,i}\right] = \underbrace{\left(-\frac{\mathsf{Cov}_t\left(R^e_{t+1,i}, M_{t+1}\right)}{\mathsf{Var}_t(M_{t+1})}\right)}_{\beta_{t,i}} \cdot \underbrace{\frac{\mathsf{Var}_t(M_{t+1})}{\mathbb{E}_t\left[M_{t+1}\right]}}_{\lambda_t}$$

where $\beta_{t,i}$ is the exposure to systematic risk and λ_t is the price of risk.

No-Arbitrage Asset Pricing

We define the SDF as the following

$$M_{t+1} = 1 \sum_{i=1}^{N} \omega_{t,i} R^e_{t+1,i} = 1 - \omega_t^\top R^e_{t+1}.$$

Since $\mathbb{E}\left[R^e_{t+1} M_{t+1}\right] = 0$ implies the SDF weights

$$\omega_t = \mathbb{E}_t \left[R^e_{t+1} (R^e_{t+1})^\top\right]^{-1} \mathbb{E}\left[R^e_{t+1}\right]$$

.

These are the portfolio weights of the coniditional mean-variance on the efficient portfolio.

No-Arbitrage Asset Pricing

Let $F_{t+1} = \omega_t^\top R_{t+1}^e$ be the portfolio tangent to the efficiency frontier. Then we can rewrite the equation as the following

$$\mathbb{E}\left[R_{t+1,i}^e\right] = \frac{\mathsf{Cov}_t\left(R_{t+1,i}^e, F_{t+1}\right)}{\mathsf{Var}_t(F_{T+1})} \cdot \mathbb{E}_t\left[F_{t+1}\right] = \beta_{t,i}\mathbb{E}_t\left[F_{t+1}\right]$$

So the no-arbitrage implies a one-factor model

$$R_{t+1,i}^e = \beta_{t,i}F_{t+1} + \epsilon_{t+1,i}$$

where $\mathbb{E}\left[\epsilon_{t+1,i}\right], \mathsf{Cov}_t\left(F_{t+1}, \epsilon_{t+1,i}\right) = 0.$

Generative Adversarial Methods

Generative Adversarial Methods

Let $g(\cdot) : \mathbb{R}^p \times \mathbb{R}^D$, where $I_t \times I_{t,i} \in \mathbb{R}^p \times \mathbb{R}^q$ denotes all the variables in the information set at time t and D is the number of moment conditions. Then the conditional no-arbitrage moment condition can be defined as the following:

$$\mathbb{E}\left[M_{t+1} R^e_{t+1,i} g(I_t, I_{t,i})\right] = 0,$$

where I_t denotes p macroeconomic conditioning variables that are not asset specific, and $I_{t,i}$ are q firm-specific characteristics.

Generative Adversarial Methods

The paper uses a theoritical adversarial approach to select the moment conditions that lead to the largest mispricings:

$$\min_{\omega} \max_{g} \frac{1}{N} \sum_{j=1}^{N} \left\| \mathbb{E}\left[\left(1 - \sum_{i=1}^{N} \omega(I_t, I_{t,i}) R^e_{t+1,i} \right) R^e_{t+1,j} g(I_t, I_{t,j}) \right] \right\|^2$$

where the function ω and g are trained in a neural network.

Generative Adversarial Method

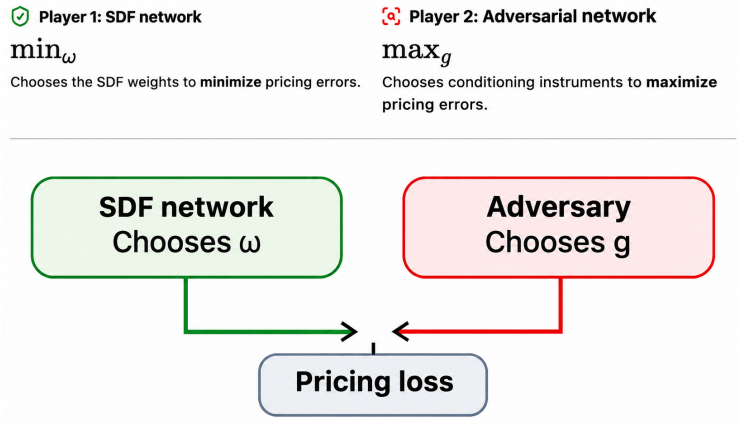


Figure 1: Adversarial Pricing Loss Diagram

Generative Adversarial Method

The empirical loss function minimizes the weighted sample moments and specifics how the SDF network measures pricing errors.

$$L\left(\omega \mid \hat{g}, I_t, I_{t,i}\right) = \frac{1}{N}\sum_{i=1}^{N}\frac{T_i}{T}\left\|\frac{1}{T_i}\sum_{t\in T_i} M_{t+1}R^e_{t+1,i}\hat{g}(I_t, I_{t,i})\right\|^2$$

Model Architecture

GAN Model Architecture

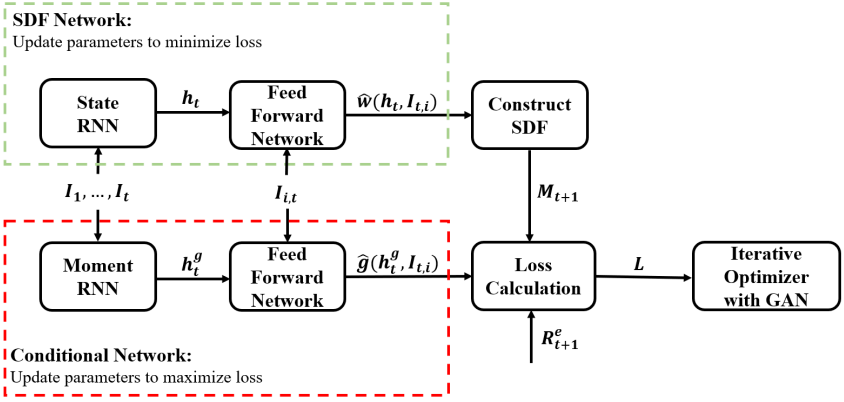


Figure 1 from Chen, Pelger, and Zhu, Deep Learning in Asset Pricing (SSRN 3350138).

RNN with LSTM

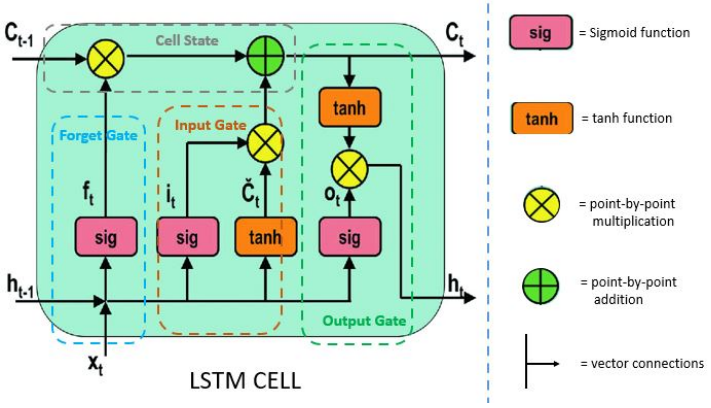


Figure 2: Long short-term memory

LSTM: General Mathematical Framework

1. Input Layer

$$x_t \in \mathbb{R}^p, \quad h_{t-1}, c_{t-1} \in \mathbb{R}^d$$
$$u_t = [h_{t-1}; x_t]$$

2. LSTM Gates

$$f_t = \sigma(W_f u_t + b_f),$$
$$i_t = \sigma(W_i u_t + b_i),$$
$$o_t = \sigma(W_o u_t + b_o)$$

3. Memory Update

$$\tilde{c}_t = \tanh(W_c u_t + b_c),$$
$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

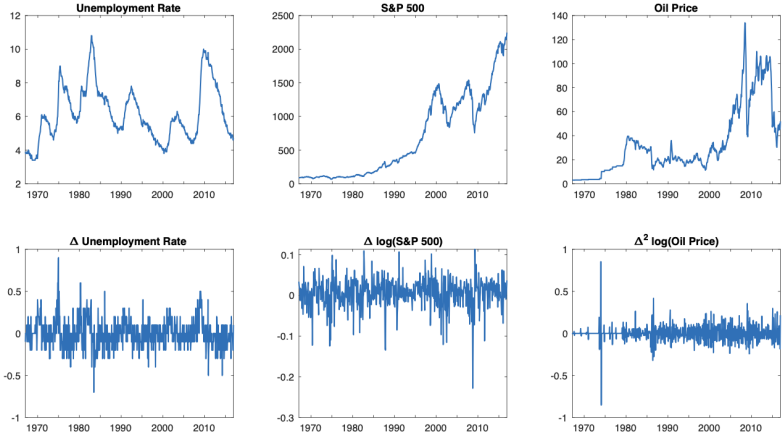
4. Hidden State Output

$$\boxed{h_t = o_t \odot \tanh(c_t)}$$

Notation

- x_t: Input vector
- h_t: Hidden state
- c_t: Memory cell
- f_t: Forget gate
- i_t: Input gate
- o_t: Output gate
- W, b: Weights and biases
- σ: Sigmoid function
- $\odot$: Element-wise product

LSTM - Examples of Macroeconomic Indicators



This figure shows examples of macroeconomic time series with standard transformations proposed by McCracken and Ng (2016).

Figure 3: Examples of Macroeconomic Variables

LSTM Architecture Comparison

Component	SDF Network	Conditional Network
Input dimension p	178	178
Hidden-state dimension d	4	32
LSTM output	$h_t \in \mathbb{R}^4$	$h_t^g \in \mathbb{R}^{32}$
Next component	SDF FFN	Conditional FFN
Training objective	Minimize loss	Maximize loss

Table 1: Comparison of the LSTM architectures used in the SDF and conditional networks of Chen, Pelger, and Zhu (2021).

Feed Forward Network

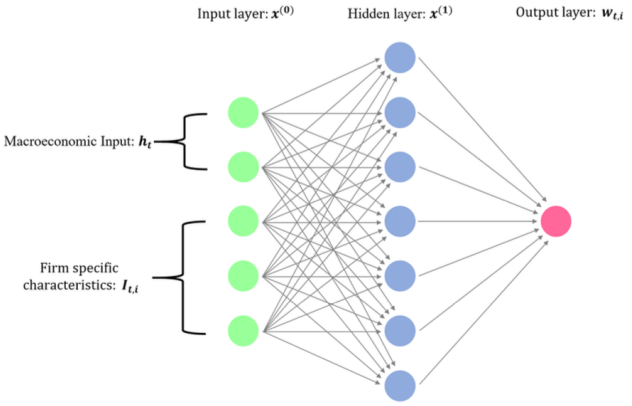


Figure 4: Feedfoward Network with Single Hidden Layer

Feedforward Neural Network: General Framework

1. Input Layer

$$x^{(0)} = x \in \mathbb{R}^{d_0}$$

2. Hidden Layers $(\ell = 1, \ldots, L)$

$$\boxed{x^{(\ell)} = \sigma\left(W^{(\ell)} x^{(\ell-1)} + b^{(\ell)}\right)}$$

3. Output Layer

$$\boxed{\hat{y} = \phi\left(W^{(L+1)} x^{(L)} + b^{(L+1)}\right)}$$

4. Parameter Estimation

$$\hat{\theta} = \arg\min_{\theta} \mathcal{L}(\theta)$$

Notation

- $W^{(\ell)}$: Weight matrix
- $b^{(\ell)}$: Bias vector
- σ: Hidden activation (ReLU)
- ϕ: Output activation
- L: Number of hidden layers

Feedforward Neural Network Comparison

Component	SDF Network	Conditional Network
Input dimension d_0	50	78
Macroeconomic states	4	32
Firm characteristics	46	46
Hidden layers L	2	0
Neurons per hidden layer	64	N/A
Output dimension	1	8
Network output	$\omega_{t,i}$	$g_{t,i}$
Optimization	Minimize loss	Maximize loss

Table 2: Comparison of the feedforward neural network architectures used in the SDF and conditional networks of Chen, Pelger, and Zhu (2021).

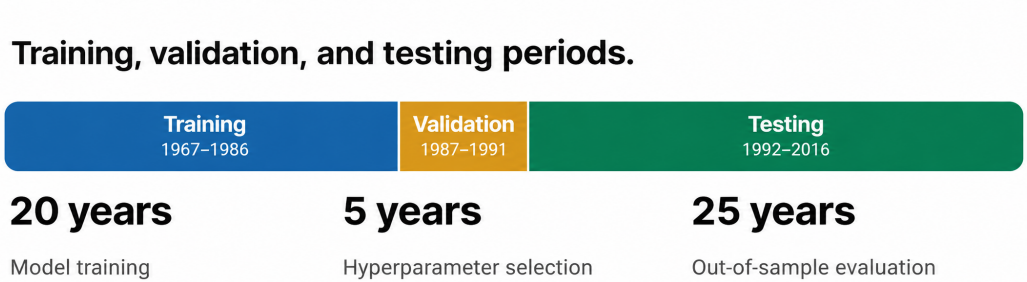
Data

Data Overview

Data	Source	Description
Stock returns	CRSP	Monthly U.S. equity returns
Firm characteristics	CRSP/Compustat	46 stock-specific characteristics
Marco variables	FRED-MD and other sources	178 monthly time series
Risk-free rate	Kenneth French Data Library	One-month Treasury bill rate

Table 3: Overview of the datasets used in Chen, Pelger, and Zhu (2021). The sample covers U.S. equities from January 1967 to December 2016.

Time Split Training



Performance

GAN Illustration

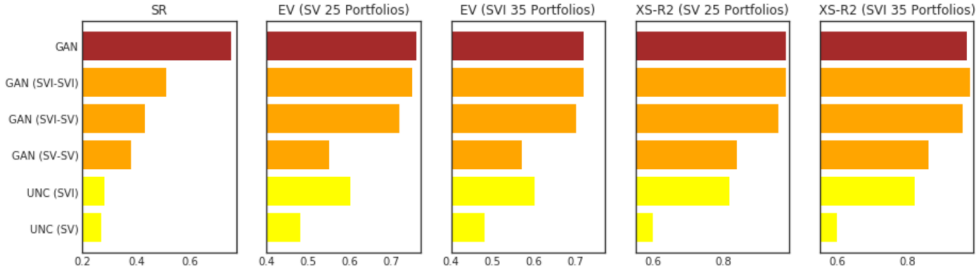


Figure 5: GAN Illustration, from Chen, Pelger, and Zhu (2021), demonstrates how adversarially selecting test assets improves the performance of a stochastic discount factor (SDF) compared with estimating an SDF using unconditional pricing restrictions.

GAN Performance

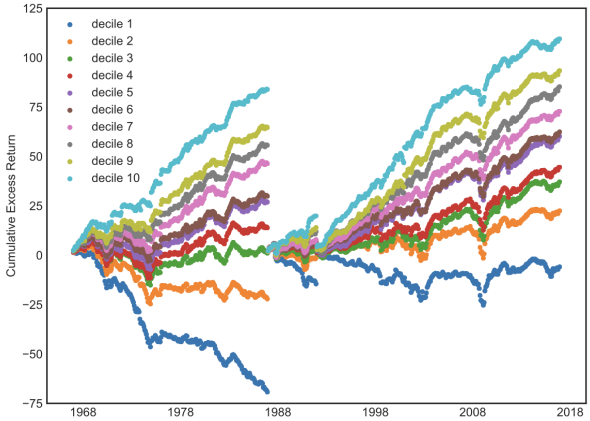


Figure 6: Cumulative Excess Returns of Decile Sorted Portfolios with GAN

Pricing Errors

	Average Returns		Market-Rf				Fama-French 3				Fama-French 5			
	Full	Test	Full		Test		Full		Test		Full		Test	
Decile			α	t	α	t	α	t	α	t	α	t	α	t
1	-0.12	-0.02	-0.19	-8.92	-0.11	-3.43	-0.21	-12.77	-0.13	-5.01	-0.20	-11.99	-0.12	-4.35
2	-0.00	0.05	-0.07	-4.99	-0.04	-1.56	-0.09	-8.79	-0.05	-3.22	-0.09	-8.29	-0.05	-2.68
3	0.04	0.08	-0.02	-2.01	-0.00	-0.16	-0.04	-5.18	-0.02	-1.40	-0.04	-4.87	-0.01	-1.05
4	0.07	0.09	-0.00	-0.03	0.01	0.68	-0.02	-2.30	-0.00	-0.35	-0.02	-2.86	-0.01	-0.54
5	0.10	0.12	0.03	2.75	0.04	2.50	0.01	2.08	0.03	2.46	0.01	1.36	0.03	2.17
6	0.11	0.12	0.04	3.16	0.05	2.77	0.02	2.75	0.03	2.85	0.01	1.51	0.02	2.20
7	0.14	0.15	0.07	5.62	0.07	3.92	0.05	6.61	0.05	4.39	0.04	5.16	0.04	3.41
8	0.18	0.18	0.11	7.41	0.10	5.12	0.08	9.32	0.08	5.83	0.07	8.05	0.07	4.86
9	0.22	0.21	0.15	7.83	0.13	5.37	0.11	9.16	0.11	5.71	0.11	8.58	0.11	5.39
10	0.37	0.37	0.29	9.22	0.27	6.05	0.24	10.03	0.25	6.27	0.25	10.43	0.27	6.59
10-1	0.48	0.39	0.47	18.93	0.38	10.29	0.45	18.50	0.38	10.14	0.46	18.13	0.39	9.96
GRS Asset Pricing Test			GRS	p	GRS	p	GRS	p	GRS	p	GRS	p	GRS	p
			42.23	0.00	11.58	0.00	39.72	0.00	11.25	0.00	37.64	0.00	10.75	0.00

Figure 7: Time Series Pricing Errors for β-Sorted Portfolios

Explained Variation

	Explained Variation			Cross-Sectional R^2				Explained Variation			Cross-Sectional R^2		
Charact.	EN	FFN	GAN	EN	FFN	GAN	Charact.	EN	FFN	GAN	EN	FFN	GAN
ST_REV	0.43	0.58	0.70	0.45	0.79	0.94	Q	0.68	0.70	0.78	0.97	0.92	0.96
SUV	0.42	0.75	0.83	0.64	0.97	0.99	Investment	0.54	0.65	0.75	0.91	0.94	0.98
r12_2	0.26	0.27	0.54	0.66	0.71	0.93	PM	0.52	0.42	0.68	0.90	0.86	0.93
NOA	0.58	0.69	0.78	0.94	0.96	0.95	DPI2A	0.57	0.70	0.78	0.90	0.95	0.97
SGA2S	0.52	0.63	0.73	0.93	0.95	0.96	ROE	0.59	0.56	0.76	0.91	0.86	0.97
LME	0.83	0.78	0.86	0.96	0.95	0.97	S2P	0.69	0.79	0.82	0.98	0.98	0.97
RNA	0.50	0.48	0.69	0.93	0.87	0.96	FC2Y	0.56	0.71	0.76	0.91	0.94	0.95
LTurnover	0.52	0.57	0.68	0.88	0.89	0.96	AC	0.63	0.79	0.82	0.96	0.98	0.98
Lev	0.52	0.63	0.73	0.90	0.92	0.95	CTO	0.59	0.73	0.79	0.92	0.96	0.97
Resid_Var	0.52	0.27	0.65	0.84	0.73	0.97	LT_Rev	0.60	0.59	0.72	0.93	0.85	0.94
ROA	0.51	0.44	0.70	0.92	0.93	0.98	OP	0.56	0.48	0.74	0.97	0.88	0.98
E2P	0.48	0.44	0.67	0.86	0.80	0.95	PROF	0.58	0.62	0.76	0.91	0.98	0.95
D2P	0.47	0.51	0.72	0.82	0.85	0.94	IdioVol	0.43	0.27	0.66	0.79	0.72	0.97
Spread	0.49	0.32	0.60	0.76	0.71	0.92	r12_7	0.37	0.42	0.66	0.84	0.86	0.93
CF2P	0.46	0.47	0.66	0.90	0.89	0.99	Beta	0.45	0.46	0.62	0.83	0.87	0.97
BEME	0.70	0.75	0.82	0.97	0.94	0.98	OA	0.65	0.78	0.83	0.88	0.92	0.93
Variance	0.48	0.27	0.61	0.74	0.72	0.90	ATO	0.58	0.70	0.77	0.96	0.98	0.99
D2A	0.57	0.71	0.78	0.96	0.96	0.97	MktBeta	0.44	0.44	0.64	0.81	0.85	0.97
PCM	0.66	0.79	0.82	0.97	0.98	0.99	OL	0.60	0.73	0.78	0.95	0.97	0.97
A2ME	0.72	0.79	0.83	0.97	0.96	0.98	C	0.51	0.65	0.73	0.90	0.93	0.95
AT	0.77	0.70	0.83	0.77	0.89	0.92	r36_13	0.54	0.53	0.69	0.92	0.82	0.93
Rel2High	0.46	0.33	0.60	0.90	0.83	0.97	NI	0.51	0.60	0.75	0.88	0.96	0.99
CF	0.61	0.64	0.78	0.89	0.85	0.96	r2_1	0.51	0.52	0.69	0.87	0.90	0.95

Figure 8: Explained Variation and Pricing Errors for Decile Sorted Portfolios